

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: ECE 3123/ECE_3123 COMMUNICATION NETWORKS
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) For the generator polynomial 110011 and the data bits 11100011, find the CRC and the transmitted string. (4)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 1B) Four channels, two with a bit rate of 200kbps and two with a bit rate of 150 kbps, are to be multiplexed using multiple slots TDM with no synchronization bits. Find: i) Frame size ii) Frame rate iii) Data rate. (3)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 1C) Represent the bit stream 11010100 using the following line encoding techniques: i) Manchester ii) Differential Manchester iii) RZ iv) NRZ-I v) NRZ vi) Bipolar (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Draw the flow diagram of CSMA/CA and explain its features. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Illustrate a priority-based token ring with suitable example. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2C) In a 10 Base5 cable, find: (3)
i) Time taken by a bit to travel from the beginning to the end of the network. Ignore any propagation delay in the equipment.
ii) Maximum time it takes for a sender to detect a collision.
iii) Minimum size of an Ethernet frame for collision detection.
iv) The bit length in the medium.

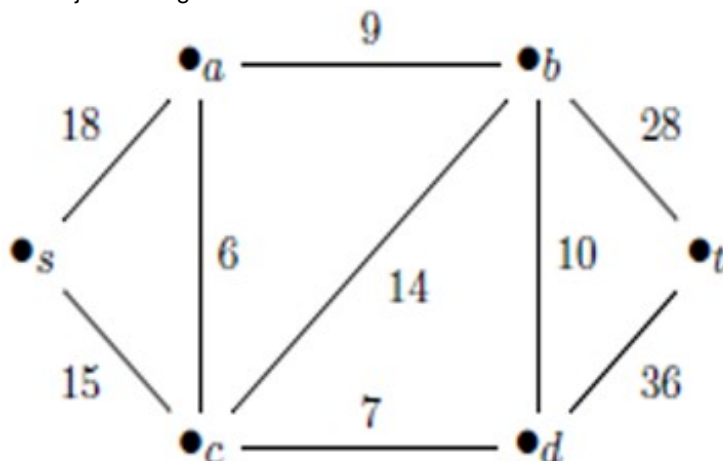
Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 3A) If the IPv4 packet is received with the following header, verify for errors if any, using an appropriate error control technique. (4)

4	5	0	(28) ₁₀	
1			0	0
4	(17) ₁₀		8BB1	
10.12.14.5				
12.6.7.9				

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 3B) Use Dijkstra's algorithm to find the shortest distance between nodes 's' and 't'. (3)



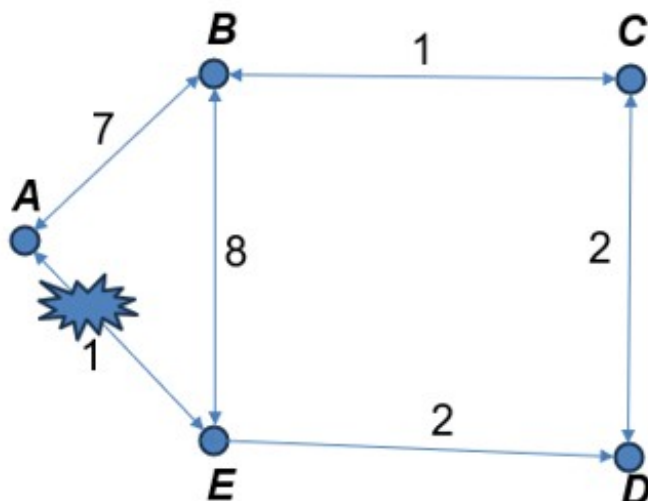
NOTE: Tabulate the routing updates.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 3C) Discuss the congestion prevention policies at network layer and transport layer. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) Form routing table at Node A, before and after link breaks between A and E using Bellman-Ford routing algorithm. (4)



Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) MAHE is granted with a block of addresses and one of the addresses is 160.124.228.200/19. MAHE needs to allocate these addresses to its three Institutes Engineering, Management, and Medical as follows: (3)
- i) Engineering has 24 departments and each department needs 115 addresses
 - ii) Management has 20 departments and each department needs 56 addresses
 - iii) Medical has 10 departments and each needs 28 addresses.
- Design the subblocks for each institute and mention the range of addresses still available after

these allocations.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

4C) Draw the IPV4 header structure, and discuss the details of each field. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5A) With neat diagram, explain the functions of each field in the TCP header structure. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

5B) Illustrate with neat diagrams, the sequence of TCP connection establishment and termination. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

5C) Write the steps that include the functioning of DNS protocol. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3123:COMMUNICATION NETWORKS

#	Course Outcome	Description	Marks
1	2	CO 2	6
2	3	CO 3	14
3	4	CO 4	20
4	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_3123:COMMUNICATION NETWORKS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	3(6%)
3	Application	6	20(40%)
4	Analysis	4	14(28%)
5	Create	4	13(26%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)